

Transportation via pipelines

Pipelines are used for transportation where simultaneous distribution of gaseous hydrogen to multiple points or intercity transmission is required. Since hydrogen gas is highly reactive, the pipeline must be able to resist interacting with the gas. For transporting hydrogen via the existing gas pipeline system, several methods are available, such as injecting and blending the hydrogen with the natural gas (injecting hydrogen at specific concentrations to reduce carbon content); converting the existing infrastructure to develop a dedicated hydrogen network; constructing a new hydrogen infrastructure and combining the hydrogen with the captured CO₂ to produce e-methane and injecting it in the gas network (methanation).

The conversion of the existing pipeline for carrying a blend of natural gas and hydrogen may require only modest modification in the pipeline, whereas substantial modification may be required for carrying the pure hydrogen via the existing pipeline. Repurposed natural gas pipeline is the most cost-efficient method for long-distance transportation of hydrogen. The construction of a new pipeline may be more cost-effective where the demand for hydrogen exceeds the carrying capacity of the existing pipeline.

Transporting hydrogen in liquefied form

Hydrogen switches to liquid phase at -253°C (-423.4°F) and the density increases by a factor of around 800. The liquid hydrogen is stored and transported in specialist super-insulated vessels and cryogenic tanker trucks using sophisticated, cost-intensive infrastructure, such as refrigeration and liquefaction plants. At the distribution site, hydrogen is vapourised to a high-pressure gaseous form. Transportation of liquid hydrogen is the safest option, environmentally.

Transporting hydrogen as ammonia

Hydrogen forms ammonia when it combines with nitrogen. However, the reconversion or cracking of ammonia back into nitrogen and hydrogen is relatively an inefficient process. Being cheaper, ammonia is trending as a promising hydrogen energy carrier. Moreover, it is much more energy dense by volume than the compressed or liquefied hydrogen and easier to store, being a liquid at room temperature under a low pressure of 10 bar. Another advantage is that it is traded internationally and capable of being used as fuel for the ships themselves.

However, ammonia is harmful and has safety risks in shipping. It is water-seeking and toxic in nature to human beings and marine life. When ammonia spills in water, it reacts with the condensation in the air to form a heavy toxic mist that does not dissipate easily. These spills pose the risk of causing explosions or fires. Hence risk mitigation measures are necessary to prevent leakages which are hazardous and can pollute the environment.

Transporting hydrogen using liquid organic hydrogen carriers

Liquid organic hydrogen carriers (LOHC) are organic compounds that absorb and release hydrogen through chemical reactions and can be used as storage media for long distance transportation of hydrogen. The transportation is carried out in a two-step cycle wherein the first step involves loading/storage of hydrogen (hydrogenation) into the LOHC molecules while the second step is unloading/releasing hydrogen (de-hydrogenation) from the LOHC. The technology has great potential for efficient and stable storage and



Fig. 7: A liquid hydrogen tanker (Source: www.energy.gov/eere)

safe and economical large-scale trans-oceanic transportation of hydrogen.

Transportation of hydrogen by road, rail, and sea

Generally, gaseous and liquid hydrogen, as well as ammonia, are transported via trucks and rail for short distances to create a virtual pipeline. Transportation of hydrogen as ammonia via trucks and rail is cost-effective at larger scales only, because of the high energy requirement for hydrogen-ammonia conversion.

Transporting liquid hydrogen or ammonia across long distances via ships is more cost-effective and is preferred for locations where hydrogen is used for decarbonisation, because it can be used directly as fuel or feedstock without the need for reconversion. Transportation of hydrogen via ships in gaseous form tends to be costlier because of the lower energy density.

The challenges

GH presents significant challenges, such as high costs, low efficiency of processes and non-availability of adequate infrastructure. Let us look at the key challenges.

Scaling up production. The production of GH is relatively at a low level compared to the global demand and needs to be scaled up significantly. This requires substantial investment and commitment from the governments as well as the industry.